That's enough for a start. Now let's use VMPK to learn about MIDI connectivity. This knowledge is useful while connecting your device with giants like Rosegarden. Also, it is a better idea to check the connectivity with simple software like VMPK when you get stuck with Rosegarden or something like that.

Learning physical MIDI connectivity using VMPK

Connecting an external device can be simpler than making connections between software applications! You simply connect the external device to yours and to the computer using a USB cable or MIDI-to-USB adapter, load the MIDI program and your device should be listed in the set of available ports.

In the case of VMPK, the ports are listed in *Edit->Connections.* Select your device from the *Input MIDI Connection* list and your software synthesiser (TiMidity++, FluidSynth, etc) from the *Output MIDI Connection* list. This enables the computer to render the music that you play on your keyboard. The same method can be used to enter notes into a scorewriter program (in which case, you have to set up the scorewriter, not VMPK).

Swapping the MIDI's in and out connections will enable VMPK to control your device. This means that what you play on VMPK will be rendered by your external MIDI device (if it is a synthesiser and not just a MIDI controller).

Also, refer to the manual of your device before making connections.

MuseScore to write and render sheet music

If you like the traditional pencil-and-paper way to write music, but still want an easy-to-use app to engrave and render your work, install MuseScore.

MuseScore is a full-fledged scorewriter and rendering app that is free software and cross-platform. You can prepare music sheets to print or to render as an audio file. Yes, this is the app to get a digital orchestra on your computer!

You can install it by using the command *sudo apt-get install musescore* (or 'mscore', sometimes) or via your preferred graphical package manager like Ubuntu Software Centre. It is also available in many other forms including the portable Appimage format, which you can get from *musescore.org.*

What makes MuseScore attractive

Being free software and cross-platform is enough for an application to be attractive. But MuseScore goes beyond this.

1. **Full-fledged score writing**: With unlimited staves and almost all symbols that you use in a score, MuseScore is capable of assisting you in the production of any type of music sheet. It seems to have a little bit of intelligence too. You will be thrilled when it automatically repositions staves, measures, lyrics and signatures as you make changes to the score. MuseScore 2 has a feature called 'linked part facility', which automatically gets the changes you make to a part applied to the other parts linked to it.

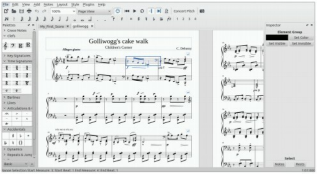

Figure 3: MuseScore

2. **Live and realistic playback:** People don't consider typesetting the score as the main objective of a score writer application. Rather, they focus on live playback which helps to verify and give a 'feel' of the work.

Like many music production applications, MuseScore supports JACK MIDI. But it doesn't stop there. It has its own built-in synthesiser, which provides audio controls like reverb and chorus, and even tries to support multiple soundfonts. Instrument-level control is also available.

MuseScore 2 goes a step further to interpret music symbols like a crescendo and tremolo.

3. **Support for multiple file formats:** Being cross-platform alone doesn't make an application universal. It should support different file formats also.
   Formats that MuseScore can import and export include MusicXML and MIDI, which makes it interoperable with most mainstream music production applications. Audio can be exported to WAV, FLAC, MP3 and OGG. The engraved score can be printed directly or exported to PDF, PS, SVG or PNG.
4. **Accessibility:** One important feature that attracts keyboard lovers to MuseScore is the accessibility support it provides. It recognises a lot of keyboard short cuts and supports direct keyboard note input. This enables even those who're visually challenged to produce scores. Also, it has built-in support for multiple screen readers.
5. **Portability and the Appimage version:** MuseScore is available as a portable application—it can be stored on a removable device and run on any system. This is made possible in GNU/Linux with the Appimage format, which enables cross-distro software distribution, so you can avoid installing the software and using root/sudo privileges.
6. **Online community:** *MuseScore.com* is a website where people can share the sheets created using MuseScore. It has a built-in player that resembles the MuseScore interface. The site also features the 'VideoScore' facility that enables the score to be linked and synchronised with a YouTube video. But you need a *pro* account to upload more than five scores.*MuseScore.org* provides the software download, and hosts documentation and forums